Not Dead Yet: Options Trading Floors

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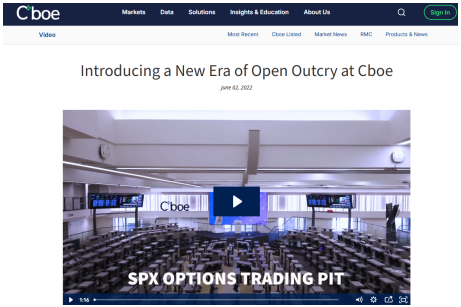
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The Microstructure Exchange – Online Seminar

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Cboe Keeps Investing in Its 1973 Trading Floor



Cboe and CNBC Launch Multi-Year Collaboration to Deliver Daily Live Market Coverage from Chicago Trading Floor

NEWS PROVIDED BY
Cboe Global Markets, Inc.
Mar 31, 2026, 10:00 ET

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- Cboe electronic trading: small retail orders (RAES) in 1985, an electronic book in 1987, the Hybrid system in 2003, and AIM auctions in 2006
- June 2022: new 40,000 sq ft floor, ten pits
- 2026: CNBC broadcasts daily from the floor
- **Why invest in open outcry today?**

Human Intermediation Survived Only in Options

Equities

- The NYSE floor has about 3% of volume, mostly opening and closing auctions

Futures

- CME closed nearly all futures pits in 2015

Options

- The floor trades **over 20%** of dollar volume
- Almost all of it on **Cboe**

⏩ By Exchange



Chicago Sun-Times, "At Chicago's equity options market, 'open outcry' cranks up the volume," 2022

Research Question

Options trade only **on exchanges**. All 17 run electronic limit order books, and 11 also run electronic auctions.

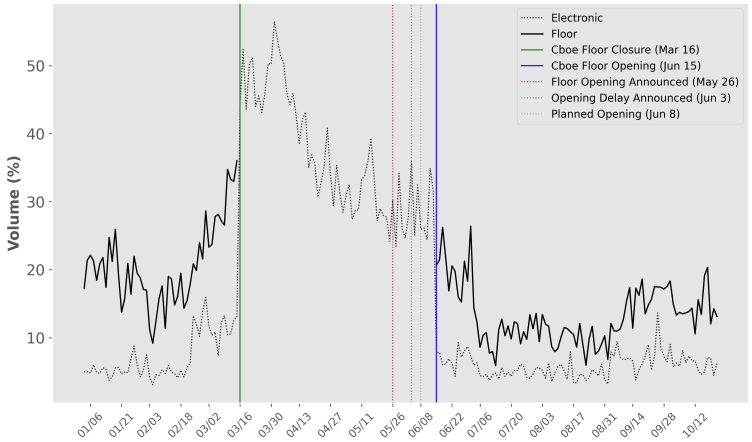
What does physical floor trading still contribute in such automated markets?

Specifically:

- What trades on the floor?
- What does floor execution cost?
- What would be lost without it?

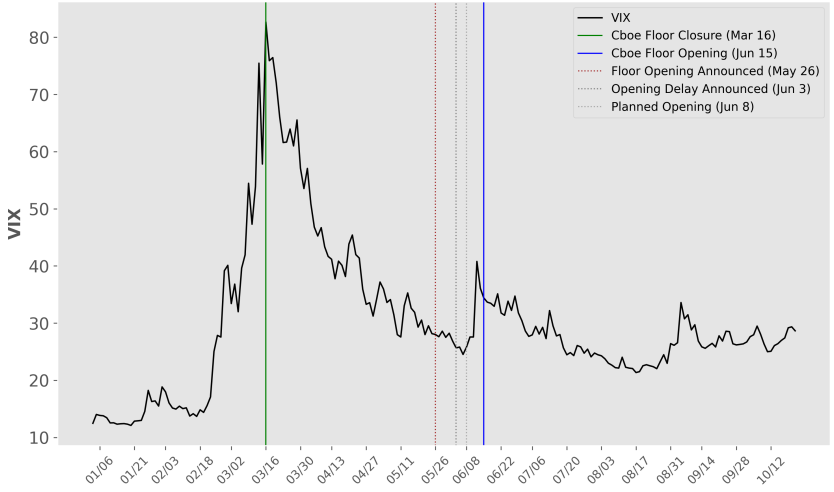
The Floor is SPX Complex

A **complex** order combines two or more option contracts (call or put) into a single trade ⏩ Example



- Floor ⏩ SPX Complex: **16%** of OPRA dollar volume; SPX: **97%** of Cboe floor

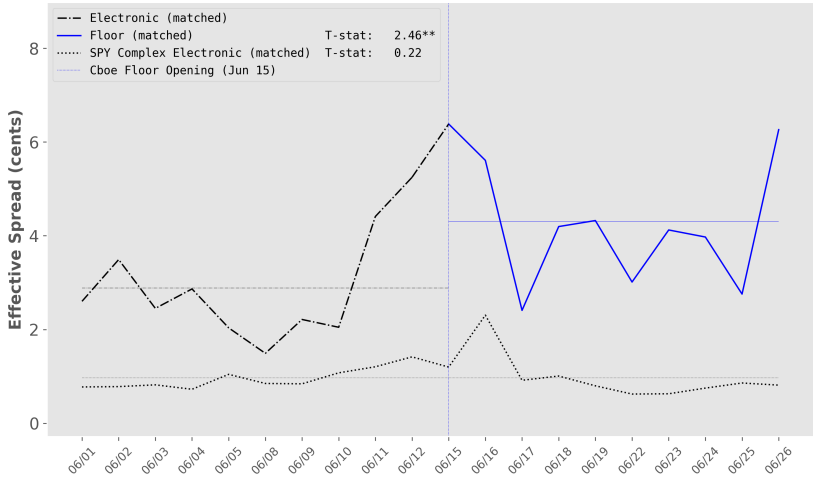
Volatility During Covid-19



- Volatility was low at the opening and we control for VIX

Other Details

Execution Costs Jump When the Floor Resumes



- Matched SPX spreads rise from **2.9¢ to 4.3¢** at the opening; SPY spreads do not

⏩ MM Revenues

This Paper in the Literature

1. **Humans versus machines in trading**
   - Venkataraman (2001); Barclay, Hendershott & McCormick (2003); more recently: Brogaard, Ringgenberg & Roesch (2025); Hu & Murphy (2026)
   - We add a clean mechanism experiment

2. **Options execution costs**
   - Kaeck, van Kervel & Seeger (2022): SPX spreads far exceed price impact
   - Li, Musto & Pearson (2023): complex orders execute at much lower cost than simple orders
   - Our results: adverse selection costs are low and confirm that complex orders execute at narrow spreads

3. **Options auction design**
   - Bryzgalova, Pavlova & Sikorskaya (2023); Ernst & Spatt (2026); Anand & Muravyev (2025); Hendershott, Khan & Riordan (2026)
   - We show that design affects what trades in auction form

4. **Block trading and upstairs markets**
   - Seppi (1990); Grossman (1992); Madhavan & Cheng (1997); Bessembinder & Venkataraman (2004)
   - We show that a floor can provide some benefits of upstairs trading

Questions So Far?

What you have seen

- Floor survives in SPX complex trades
- Volatility spiked during the closure
- Matched execution costs rose by half when the floor reopened

Coming next

- **Result 1**: what trades on the floor
- **Result 2**: what the floor charges
- **Result 3**: what would be lost without it

Data

Cboe Trade-by-Trade Execution Detail (DataShop)

- Every Cboe execution in 2020, one record each for buyer and seller
- Legs share a complex execution ID (electronic) or an agreement time (floor)
- Trades are grouped into complex orders ⏩ Grouping Example

OPRA: market-wide volume shares and the SPY placebo

LSEG Tick History

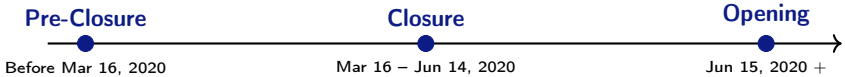
- Option quotes at the agreement time (floor) or execution time (electronic)
- E-mini S&P 500 futures quotes, matched to each option trade at its execution time (electronic, microsecond) or agreement time (floor, millisecond)

Sample: January 2 – October 21, 2020

A Clean Laboratory

Why SPX options are ideal

- All options trade on exchanges, so there are no dark venues
- SPX trades **only** on Cboe, so trading cannot move to another venue



- Floor trading
- **No AIM auction**
- Electronic CLOB and COA

- **No floor trading**
- AIM auction
- Electronic CLOB and COA

- Floor trading
- **No AIM auction**
- Electronic CLOB and COA

- During the closure, AIM was activated also for SPX complex orders ▸ Auction Details
- The opening was delayed by civil unrest, not by market conditions ▸ Rule Changes

Two Characteristics Predict Floor Routing: Size and Legs

Pr(Floor Execution)	Size only	Legs only	Both	Full	
Log Trade Size	2.01^{***}		1.82^{***}	1.80^{***}	1.83^{***}
	(152.81)		(132.73)	(115.65)	(111.37)
Log Legs		3.64^{***}	2.13^{***}	1.60^{***}	3.34^{***}
		(98.17)	(51.25)	(38.12)	(8.05)
Greeks, IV, Expiry, SPXW	No	No	No	Yes	Yes
Strategy FE	No	No	No	No	Yes
Pseudo R^2	**0.64**	0.32	**0.70**	0.75	0.80

Up to 1,698,094 opening-period complex SPX orders, 27,839 on the floor; z-statistics from date-clustered standard errors

- Size alone gives a pseudo-R^2 of **0.64**, size and legs **0.70**
- Doubling size adds 0.51 pp, doubling legs 0.93 pp (base rate 1.6%)

Marginal Effects
Strategies

Average Costs Fall at the Opening: Composition or Mechanism?

SPX Complex Trades, Daily Means	Pre-Closure	Closure	Opening
Volume (% of SPX $ volume)	86.43	87.86	87.49
Effective Spread (EW, ¢)	**1.93**	**2.34**	**1.82**
Effective Spread (VW, ¢)	5.03	6.51	3.31
MM Revenue (EW, ¢)	1.52	1.98	1.43
Hedge Cost (EW, ¢)	0.39	0.34	0.37

$$\textbf{Effective Spread} = \text{Direction} \times (\text{Net Price} - \text{Midpoint})$$

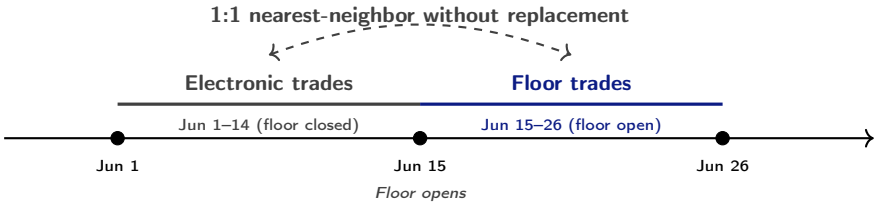
$$\textbf{Hedge Cost} = |\Delta| \times \tfrac{1}{2}\,\text{Futures Spread}$$

$$\textbf{MM Revenue} = \text{Effective Spread} - \text{Hedge Cost}$$

- SPX complex spread rise in the closure and fall at the opening
- Does the floor change costs, or only which trades execute?

⏩ Midpoint Drift

Matching Floor Trades to Electronic Counterparts



- Matched on |delta|, |vega|, IV, size, expiry, and legs (0.25 SD calipers)
- The same design applies around the closure

Matched Trades Are Balanced, Floor Trades Pay More

	After Matching			Floor
	Floor	Electronic	Difference	Unmatched
|Delta|	0.02	0.02	0.00	0.02
Trade Size (Contracts)	341.51	339.41	2.10	1030.58
Legs	3.08	3.07	0.00	17.21
Days to expiration	25.17	25.16	0.01	117.15
Effective Spread (¢)	**4.45**	**3.15**	**1.30**	**5.46**
Hedge Cost (¢)	0.28	0.28	0.00	0.21
MM Revenue (¢)	4.17	2.87	1.30	5.25

- 36% of floor trades find a match, giving 1,177 pairs
- Matched floor trades pay **4.45¢**, their electronic matches **3.15¢**
- Unmatched floor trades average **1,031 contracts and 17 legs**

▸▸ Full Table
▸ Closure Matching

Matched Floor Trades Cost a Third More

SPX Complex	Effective Spread		MM Revenue	
	(1)	(2)	(1)	(2)
Opening	**0.98****	**1.14*****	**1.00*****	**1.15*****
	(2.43)	(3.01)	(2.60)	(2.96)
VIX	0.16***	0.10**	0.15***	0.12***
	(4.44)	(2.53)	(4.26)	(2.78)
SPY Complex ES	0.96***	0.88***	0.97***	0.93***
	(2.94)	(3.13)	(3.15)	(3.29)
Other Controls	No	Yes	No	Yes
Strategy FE	No	Yes	No	Yes
R-squared	0.01	0.11	0.01	0.09

1,177 matched pairs; Opening = 1 for the floor trade of a pair;

t-statistics from date-clustered SEs

- Matched SPX floor trades cost **1.14¢** more, 36% of the electronic spread

⏩ Closure Event ⏩ Versus AIM

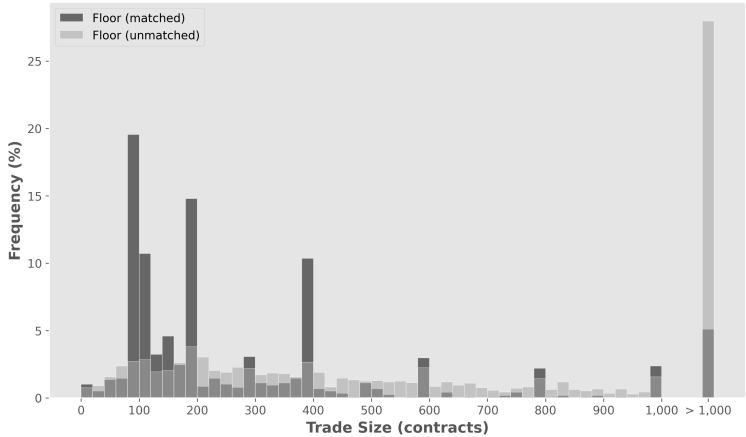
SPY Complex Trades (Placebo) Show No Change at the Opening

SPY Complex	Effective Spread		MM Revenue	
	(1)	(2)	(1)	(2)
Opening	**−0.17**	**−0.05**	**−0.17**	**−0.05**
	(−0.59)	(−0.79)	(−0.61)	(−0.79)
VIX	0.05**	0.02***	0.05**	0.02***
	(2.25)	(3.67)	(2.28)	(3.71)
Other Controls	No	Yes	No	Yes
Strategy FE	No	Yes	No	Yes
R-squared	0.01	0.46	0.01	0.46

428,627 trades; Opening = 1 from June 15, 2020;

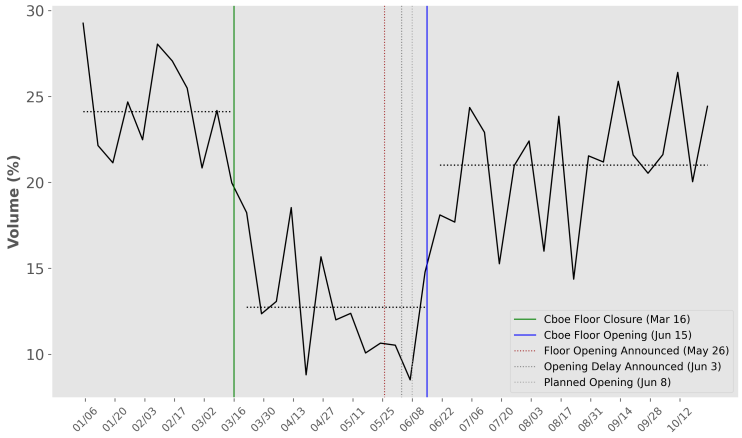
t-statistics from date-clustered SEs

Unmatched Floor Trades Are Much Larger



- 28% of unmatched floor trades exceed 1,000 contracts, 5% of matched

The Largest, Most Complex Trades Plummet Without the Floor



- Trades above the unmatched-floor median in size and legs fall from **24%** to **13%**, then recover to **21%**

Daily

By Trade Size, the Ranking Reverses

	Effective Spread		MM Revenue	
	(1)	(2)	(1)	(2)
Opening × ≤ P25	**5.86*****	**3.49*****	**5.72*****	**3.49*****
	(11.44)	(4.66)	(11.23)	(4.79)
Opening × P25–P50	1.93**	1.10	2.05**	1.14
	(2.43)	(1.30)	(2.54)	(1.35)
Opening × P50–P75	0.56	0.66	0.95*	0.66
	(1.08)	(1.35)	(1.82)	(1.36)
Opening × P75–P90	−0.88	−0.48	−0.71	−0.47
	(−0.70)	(−0.38)	(−0.56)	(−0.38)
Opening × > P90	**−4.00*****	**−4.35*****	**−4.53*****	**−4.30*****
	(−3.80)	(−4.16)	(−4.10)	(−4.16)
Controls	VIX + SPY	All	VIX + SPY	All
Strategy FE	No	Yes	No	Yes
Observations	209,117	204,969	204,969	204,969

Opening = 1 for floor trades after the opening; t-statistics from date-clustered SEs

- Small floor trades pay **3.49¢** more than electronic trades of the same size
- The largest floor trades pay **4.35¢** less; market maker revenues follow

Why Execute on the Floor? (Q&A)

Results 1–3 established

- The floor trades large multi-leg packages
- Matched floor trades cost 36% more, mostly for smaller trades
- The largest and most complex trades become much less prevalent
- The SPY effective spread is flat, and the largest SPX trades are cheaper on the floor

Result 4 coming

- Is the higher cost adverse selection?
- Or does the floor buy a service?

Is the Higher Cost Compensation for Trading Against Information?

Test: do floor trades move futures prices? If floor flow is informed, $\beta_1 + \beta_2 > 0$

$$r_{t+h}^{\text{Futures}} = \beta_1 \,\text{SignedDelta}_i + \beta_2 \,\text{Floor}_i \times \text{SignedDelta}_i + \gamma' X_i + \mu_t + \mu_s + \varepsilon_i$$

E-Mini Futures Return (bp)	10 seconds		15 seconds	
	(1)	(2)	(3)	(4)
Signed Delta ($10K)	0.43^{***}	0.44^{***}	0.42^{***}	0.43^{***}
	(3.02)	(3.11)	(3.94)	(3.99)
Floor × Signed Delta	-0.54^{**}	-0.57^{**}	-0.68^{***}	-0.71^{***}
	(−2.03)	(−2.15)	(−2.59)	(−2.68)
Floor Effect ($\beta_1 + \beta_2$)	**−0.11**	**−0.13**	**−0.26**	**−0.28**
Day FE, Controls	Yes	Yes	Yes	Yes
Strategy FE	No	Yes	No	Yes
Observations	204,969			

Signed delta normalized to $10,000 per SPX point, end-user side; date-clustered SEs

Replicating One Floor Trade Electronically: 65 Orders, 1.3 Minutes

Electronic Replication of a 90th-Percentile Floor Trade					
Target Trade Size	(contracts)		1,950		
Orders Required	(mean / median)		**65.3 / 64.0**		
Elapsed Minutes	(mean / median)		**1.3 / 1.0**		
Futures Slippage	10 s / 15 s ($)		87 / 85		
Electronic Spread Paid	($)		6,081		
Electronic Total Cost	($)		6,166–6,168		
Floor Band around Target		±1%	±2.5%	±5%	±10%
Floor Total Cost	($)	2,439	3,357	4,246	3,745
Floor Advantage	(%)	**60.4–60.5**	**45.6**	**31.1–31.2**	**39.3**

Electronic cost = spread paid + futures slippage; floor cost = mean dollar effective spread in the band

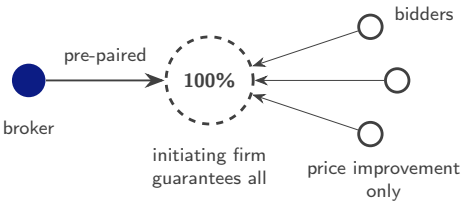
- The floor is **31–60% cheaper** for these trades

AIM Only

Is an Electronic Auction the Same as a Floor Auction? (1)

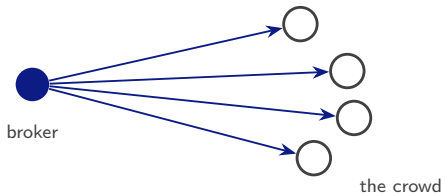
Difference 1: **pre-commitment**

Electronic auction (AIM / C-AIM)



no pre-commitment ⇒ no auction

Floor open outcry



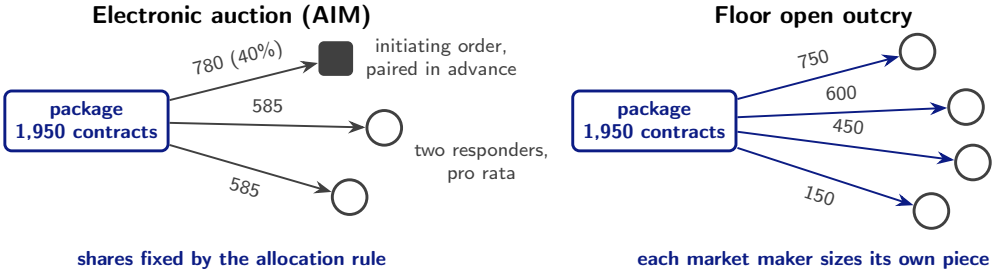
broker brings order ⇒ crowd bids, or no trade

AIM needs pre-committed capital; the floor needs none

Cboe Rules 5.37 and 5.38 (AIM, C-AIM); Rules 5.85 and 5.87 (Open Outcry)

Is an Electronic Auction the Same as a Floor Auction? (2)

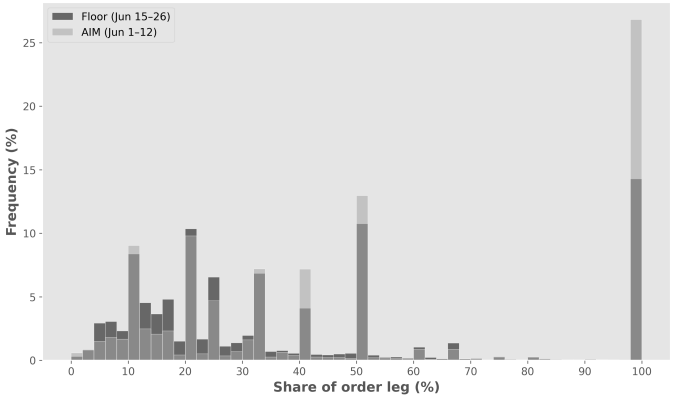
Difference 2: **flexibility**



The crowd splits the trade in many sizes; AIM often in fixed shares

- Why is there no flexible electronic auction? We leave this to future research

Floor Fills Come in Many Sizes; AIM Fills Cluster at Fixed Shares

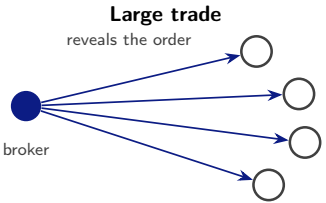


- Pooled across sizes, 27% of AIM fills cover the whole leg, against 14% on the floor
- Like AIM fills, floor fills cluster at half the leg, but less at 40%
- Conditional on execution, floor fills come in many sizes

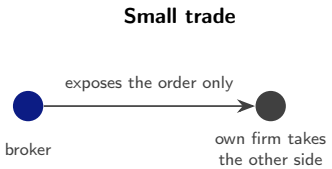
Floor Priority
Whole Legs

Is an Electronic Auction the Same as a Floor Auction? (3)

Difference 3: **screening**. Disclosure on the floor is a choice



information shared ⇒ competition for the risk



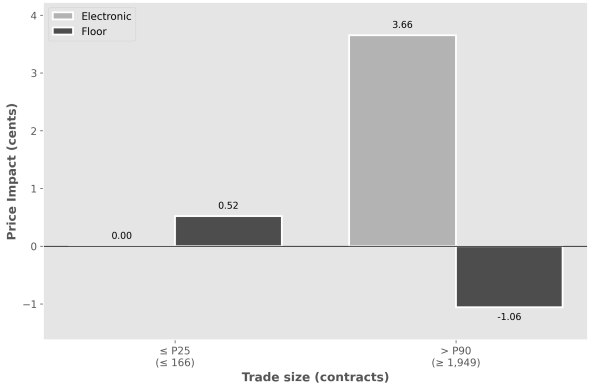
little disclosure ⇒ one counterparty takes it whole

Brokers share large orders with the crowd

- Incentives to disclose are increasing in trade size
- Large floor trades move futures less than electronic trades; small ones do not differ from electronic trades in adverse selection

⏩ Indicative Evidence

Large Floor Trades Carry Less Adverse Selection Than Electronic Trades



Price impact over one minute, in cents per unit of underlying

- Small trades: little price impact on either venue (0.00¢ and 0.52¢)
- Largest trades: 3.66¢ electronic against −1.06¢ on the floor
- Consistent with screening of large floor orders, though not conclusive Decomposition

Market Design: What Would Happen if We Kept Both Mechanisms?

- Floor and electronic execution serve **different** trades
  Floor costs 4.35¢ less for the largest trades and 3.49¢ more for the smallest
- Coexistence could lower costs if orders reach the mechanism that suits them
  Brokers choose the venue
- But the floor may need its easy trades
  Can the floor survive if AIM takes its easy trades?
- Why does Cboe not offer a more flexible electronic auction?
  Do Cboe's incentives conflict with those of market makers or customers?

Conclusion: Not Dead Yet

1. **The floor attracts complex trades, particularly the largest**
   Floor trades are **70%** of complex SPX volume, and size and legs predict routing (pseudo-R^2 **0.70**)
2. **The floor costs more, mostly for smaller trades**
   Matched floor trades cost **36%** more, but the largest floor trades pay 4.35¢ less
3. **Without the floor, the largest and most complex trades plummet**
   Their share falls from **24%** to **13%** and recovers to **21%**; rebuilding one electronically takes **65 orders** and costs **1.5 to 2.5 times** as much

The floor facilitates the most difficult trades and raises the cost of easier ones

Appendix

Floor Volume by Exchange

Return

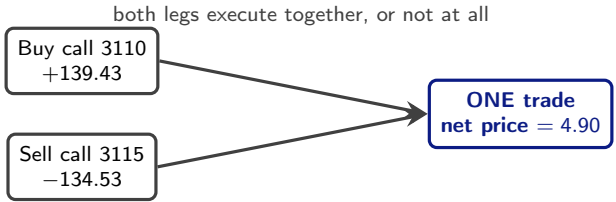
Exchange	% of OPRA $ Volume
Cboe	**16.81**
AMEX	1.95
PHLX	1.74
ARCA	1.00
BOX	0.50

- Cboe has 76% of all floor trading

Complex Trades: Many Legs, One Guaranteed Net Price

Example: long call vertical spread

Timestamp	Cond. ID	Size	Price	Ask	Bid	Type	Strike	Symbol
09:30:06.475	119	1	139.43	141.10	137.50	Call	3110	SPX
09:30:06.475	119	1	134.53	136.40	132.90	Call	3115	SPX



- The net price of 4.90 is above the 4.65 midpoint, so the initiator buys the spread
- **Effective spread** = direction × (net − midpoint) = 12.5¢ (25¢ per package) Return

The Floor Is an SPX Complex Phenomenon

Return

Volume	$B per day	% OPRA
SPX	3.96	25.83
Complex	3.39	22.32
Simple	0.57	3.51
SPX Electronic	1.51	9.52
Complex	1.01	6.36
Simple	0.50	3.16
SPX Floor	2.45	**16.31**
Complex	2.38	**15.95**
Simple	0.07	0.36

All Option Classes

- SPX is 26% of OPRA dollar volume
- The floor handles **70%** of SPX complex volume
- 97% of Cboe floor volume is SPX

Top Floor Symbols

Top Floor-Traded Symbols

Return

	2020 (%)		2021 (%)
SPX	**97.21**	**SPX**	**95.01**
SPY	0.67	SPY	1.21
AMZN	0.15	RUT	0.79
QQQ	0.13	VIX	0.79
NVDA	0.09	QQQ	0.17
IWM	0.09	TSLA	0.12
BRK.B	0.08	AMZN	0.08
BA	0.07	IWM	0.07
VXX	0.07	AAPL	0.07
ADBE	0.06	XEO	0.06

All Option Classes: Volume by Type and Venue

Return

	$ Volume		Contract Volume	
	$B	% OPRA	100,000	% OPRA
Complex	7.39	50.67	74.53	28.22
Simple	6.90	49.33	189.85	71.78
Floor	3.25	**22.00**	22.37	8.54
Electronic	11.04	78.00	242.00	91.46

- Complex trades are half of dollar volume and 28% of contracts
- The floor executes 22% of dollar volume and 8.5% of contracts

Volatility Around the Opening

Return

Daily VIX	Window	Mean
Pre-opening	Jun 1 – Jun 12	29.0
excluding Jun 11–12	Jun 1 – Jun 10	**26.6**
Post-opening	Jun 15 – Jun 26	**33.4**

On June 11, 2020, VIX rose from 27.7 to 40.8

- FOMC projections (Jun 10) and rising case counts in AZ, FL, NC, TX
- The S&P 500 fell 5.9% and the Dow 6.9%, the worst day since mid-March
- Unrelated to the floor, two trading days before the opening

- Either way, VIX is higher after the opening than before
- All regressions control for VIX

Grouping Cboe Records into Packages: Electronic

Return Floor Example

Time	Strike	Type	Side	Size	Price	Capacity	Trade Type	exec_id	complex_exec_id
10:41:10.113946	3480	C	Buy	10	0.90	Customer	Complex Add	Y1D0180LM	YF801L01M
10:41:10.113946	3480	C	Sell	10	0.90	Customer	Complex Remove	Y1D0180LM	YF801L01M
10:41:10.113946	3490	C	Sell	10	0.65	Customer	Complex Add	81YML100E	YF801L01M
10:41:10.113946	3490	C	Buy	10	0.65	Customer	Complex Remove	81YML100E	YF801L01M

- Each execution appears twice, once for each side
- Complex Remove marks the initiator, a customer selling 10 call spreads at 0.25
- One complex_exec_id marks one package, here a 1:1 vertical call spread
- Packages sharing legs, ratio, initiator, mechanism, and microsecond form one order

Grouping Cboe Records into Packages: Floor

Return Electronic Example

floor_action_ts	transact_time	Strike	Type	Side	Size	Price	Capacity	Trade Type	exec_id
14:01:02.822	14:02:57.762	2625	Put	Buy	275	1.50	Customer	PAR-C	51X4L000E
14:01:02.822	14:02:57.762	2625	Put	Sell	275	1.50	Market Maker	PAR-C	51X4L000E
14:01:02.822	14:02:57.930	2625	Put	Buy	100	1.50	Customer	PAR-C	X0L1H0054
14:01:02.822	14:02:57.930	2625	Put	Sell	100	1.50	Market Maker	PAR-C	X0L1H0054
14:01:02.822	14:02:57.762	1925	Put	Sell	275	0.15	Customer	PAR-C	XF501L004
14:01:02.822	14:02:57.762	1925	Put	Buy	275	0.15	Market Maker	PAR-C	XF501L004
14:01:02.822	14:02:57.930	1925	Put	Sell	100	0.15	Customer	PAR-C	X1I0050L4
14:01:02.822	14:02:57.930	1925	Put	Buy	100	0.15	Market Maker	PAR-C	X1I0050L4

- A customer buys 375 put spreads, filled in two executions per leg
- Exec IDs and report times differ, but one floor_action_ts marks one package
- The initiator is the side that is not a market maker
- Orders with unsigned fills are dropped (16% just before the closure, 22% just after the opening)

Electronic Auction (AIM) Mechanics

Return

- *Agency Order* and **Initiating Order** must be entered as a pair
- **Minimum net increment** is $0.05 for SPX complex AIM
- Sealed-bid auction lasting *1,000 ms* for SPX (100 ms elsewhere)
- Wholesaler enters starting price and match/reserve price
- If a responder beats the starting price but not the match price, the *Initiating Order* gets 50% (40%) at the best price against one (several) responders
- Initiating **market order guarantees** *participation* at best price
- Auction responses are permitted from all market participants
- Priority for non-initiator bidders can depend on electronic LOB quote

See Cboe Rules 5.37–5.38

Temporary Rule Changes During the Closure

Return

- AIM and C-AIM activated for SPX/SPXW in RTH for the first time
- Complex leg ratios: 3:1 limit raised to 25:1
- Appointed market makers allowed as solicited contras in single-leg AIM
- The AIM response interval was set to 1,000 ms for SPX (100 ms elsewhere)
- Orders up to 12 legs stayed electronically eligible

Cboe notice C2020031204; SEC Release No. 34-88386; all changes reversed on June 15, 2020

Logit Average Marginal Effects

Return

	(1)	(2)	(3)	(4)	(5)
Log Trade Size	0.0159^{***}		0.0115^{***}	0.0095^{***}	0.0073^{***}
Log Legs		0.0436^{***}	0.0135^{***}	0.0084^{***}	0.0134^{***}
Days-to-Expiration				-0.0003^{***}	-0.0004^{***}
Vega				0.0018^{***}	0.0010^{***}
Gamma				0.0001^{***}	-0.0000
IV				-0.0055^{***}	-0.0026^{***}
|Delta|				-0.0009	-0.0000
SPXW				-0.0128^{***}	-0.0033^{***}
Strategy FE	No	No	No	No	Yes

- In logs, doubling adds $\ln 2 \approx 0.693$ ($0.0073 \times 0.693 \approx 0.0051$, or 0.51 pp; $0.0134 \times 0.693 \approx 0.0093$, or 0.93 pp)
- The sample has 1,698,094 opening-period complex orders, 27,839 on the floor

The Largest Strategies Are Built From Combos

Return Combo Example

Strategy	|Delta| per Underlying	Hedge Cost $ per Trade	Trade Size Contracts	Trade Size Million $
Box Spread	0.01	6	249	**13.93**
Combo Roll	0.00	17	838	**7.30**
Index Combo	0.01	60	**881**	**5.77**
Combo	0.50	118	63	0.92
Put Butterfly	0.01	7	65	0.17
Calendar Put	0.01	3	16	0.10
Vertical Put Spread	0.03	7	25	0.04

Floor and electronic trades pooled; closure period excluded; shaded rows are Combo-based packages

- Index Combos average 881 contracts and $5.8M per trade; Box Spreads $13.9M

Combo-Based Share of Complex SPX $ Volume	Pre-closure	Closure	Opening
Floor	**76.71**	—	**85.01**
Electronic	1.93	**57.05**	11.04

- In the closure, combo-based packages are 57% of electronic volume, 11% after

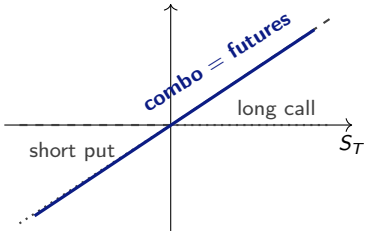
Combos Are Synthetic Futures

Example: long Combo

Timestamp	Cond. ID	Size	Price	Direction	Delta	Type	Strike	Expiration
09:31:44.625	119	6	91.29	Buy	0.54	Call	3245	2020-04-17
09:31:44.625	119	6	89.21	Sell	−0.46	Put	3245	2020-04-17

- Long call + short put, same strike and expiration
- Payoffs sum to a futures position
- Delta exposure:

$$\underbrace{1}_{\Delta} \times \underbrace{6}_{\text{contracts}} \times \underbrace{100}_{\text{multiplier}} = \$600 \text{ per point}$$



Return

Midpoints Drift Toward the Trade on Every Venue

Return

Venue	Packages	Midpoint Drift Toward the Trade (¢)		
		15 min → 1 min	10 min → 1 min	5 min → 1 min
Floor	48,946	**4.81**	4.08	2.63
AIM	292,942	4.26	3.35	2.01
COA	2,294,101	3.81	3.03	1.86
CLOB	2,300,181	1.61	1.15	0.58

- We use the last LSEG quote before agreement (floor) or execution (electronic)
- Drift is positive everywhere and largest on the floor
- An earlier benchmark would raise floor spreads the most, so the gap is not inflated

Closure Event: Matched Sample (1,447 Pairs)

Return Closure Regressions

	Before Matching			After Matching			Floor Unmatched
	Floor	Elec.	Diff	Floor	Elec.	Diff	
Panel A: Matching Covariates							
|Delta|	0.02	0.02	0.00	0.02	0.02	0.00	0.02
|Vega|	1.56	0.26	1.29	0.59	0.58	0.01	1.87
Implied Volatility	0.52	1.14	−0.62	0.67	0.70	−0.03	0.47
Trade Size (Contracts)	1245.83	52.53	1193.30	693.28	689.37	3.91	1423.10
Days-to-Expiration	82.61	42.49	40.12	45.04	47.10	−2.06	94.67
Legs	12.96	2.69	10.27	3.82	3.82	0.00	15.89
Panel B: Outcomes (cents), not matched on							
Effective Spread	16.07	4.23	11.84	**13.83**	**9.73**	**4.09**	**16.79**
Hedge Cost	0.30	0.33	−0.02	0.31	0.33	−0.02	0.30
MM Revenue	15.77	3.90	11.87	13.51	9.40	4.11	16.49

Closure Event: Matched Regressions

Return Closure Balance

	Effective Spread		MM Revenue	
	(1)	(2)	(1)	(2)
Closure	-8.35^{***}	$-\mathbf{6.33}^{***}$	-8.36^{***}	$-\mathbf{6.55}^{***}$
	(-5.71)	(-5.84)	(-5.84)	(-5.64)
VIX	0.23^{***}	0.08	0.23^{***}	0.09
	(3.28)	(1.37)	(3.38)	(1.58)
SPY Complex ES	0.49	0.99	0.50	0.83
	(0.45)	(1.04)	(0.47)	(0.85)
Other Controls	No	Yes	No	Yes
Strategy FE	No	Yes	No	Yes
R-squared	0.02	0.13	0.02	0.12

1,447 matched pairs; Closure = 1 for the electronic trade of a pair; t-statistics from date-clustered SEs

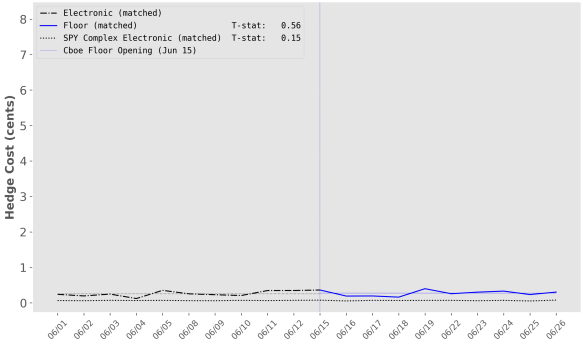
- Floor trades cost 6.33¢ more, 65% of the electronic spread
- Same sign as at the opening, in a much more volatile market

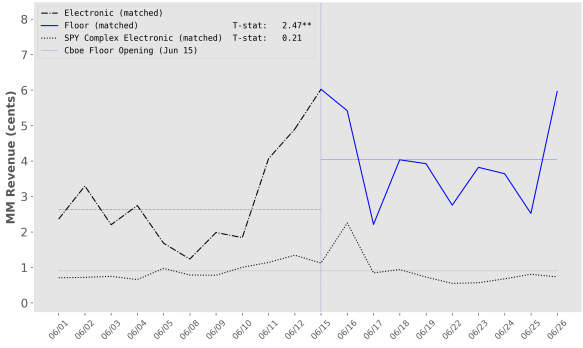
Opening Event: Full Matched-Sample Balance

Return

	Before Matching			After Matching			Floor Unmatched
	Floor	Elec.	Diff	Floor	Elec.	Diff	
Panel A: Matching Covariates							
|Delta|	0.02	0.03	−0.01	0.02	0.02	0.00	0.02
|Vega|	1.20	0.20	1.00	0.41	0.40	0.00	1.64
Implied Volatility	0.35	0.37	−0.02	0.39	0.38	0.00	0.34
Trade Size (Contracts)	783.62	36.17	747.44	341.51	339.41	2.10	1030.58
Days-to-Expiration	84.18	21.77	62.41	25.17	25.16	0.01	117.15
Legs	12.14	2.58	9.57	3.08	3.07	0.00	17.21
Panel B: Outcomes (cents), not matched on							
Effective Spread	5.10	1.97	3.13	**4.45**	**3.15**	**1.30**	**5.46**
Hedge Cost	0.24	0.36	−0.13	0.28	0.28	0.00	0.21
MM Revenue	4.86	1.60	3.26	4.17	2.87	1.30	5.25

Hedge Costs and Market Maker Revenues





- Hedge costs barely move; market maker revenues rise from 2.63¢ to 4.03¢ Return

Against AIM Alone, the Difference Doubles

Return Channel Composition

Floor versus AIM	Effective Spread		MM Revenue	
	(1)	(2)	(1)	(2)
Opening	2.45***	**2.44***	2.45***	**2.47***
	(4.93)	(6.13)	(5.19)	(6.07)
VIX	0.06*	−0.00	0.07*	0.00
	(1.73)	(−0.03)	(1.89)	(0.12)
SPY Complex ES	0.78*	0.53	0.77**	0.54
	(1.94)	(1.21)	(2.06)	(1.20)
Other Controls	No	Yes	No	Yes
Strategy FE	No	Yes	No	Yes
R-squared	0.02	0.05	0.02	0.04

1,026 matched pairs; Opening = 1 for the floor trade of a pair; t-statistics from date-clustered SEs

- AIM is the mechanism withdrawn at the opening, and the closest to the floor
- Against AIM, floor trades pay 2.44¢ more

During the Closure, AIM Absorbed the Combos

Return Floor vs AIM

	CLOB			COA			AIM
Strategy Group	Pre	Closure	Opening	Pre	Closure	Opening	Closure
Combo-based	**2.43**	**31.94**	**11.65**	1.08	4.93	9.90	**78.54**
Vertical Spreads	29.58	20.29	24.68	32.79	30.72	28.75	5.85
Butterflies and Condors	22.29	9.11	17.67	25.20	12.47	19.59	4.87
Calendars and Diagonals	16.93	15.93	26.19	17.19	28.35	25.04	4.14
Other strategies	28.79	22.74	19.79	23.75	23.52	16.73	6.55

- In the closure, combo-based packages are 79% of AIM volume and 32% of CLOB
- The flow that trades on the floor moved to AIM

Price Impact by Trade Size: Large Floor Trades Move Futures Less

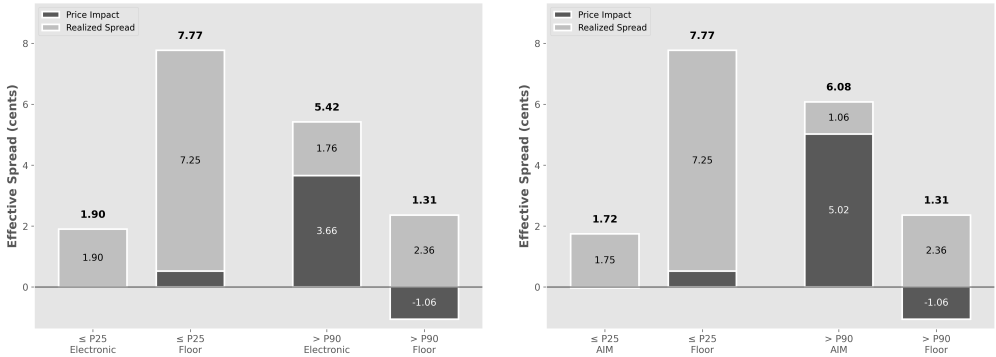
Return | Screening | Costs by Size | Spread Decomposition

E-Mini Futures Return (bp)	10 seconds		15 seconds	
	(1)	(2)	(3)	(4)
Large trades, above P90 (672 trades)				
Signed Delta ($10K)	0.26^{***}	0.31^{**}	0.27^{**}	0.24
	(4.78)	(2.20)	(2.38)	(1.63)
Floor × Signed Delta	-0.99^{**}	-1.09^{**}	-1.03^{*}	-1.09^{**}
	(−2.34)	(−2.54)	(−1.83)	(−1.98)
Small trades, at or below P25 (194,870 trades)				
Signed Delta ($10K)	−0.22	−0.08	−0.37	−0.20
	(−0.29)	(−0.11)	(−1.00)	(−0.52)
Floor × Signed Delta	−1.17	−1.04	−1.96	−1.84
	(−0.64)	(−0.59)	(−1.04)	(−0.97)
Day FE, Controls	Yes	Yes	Yes	Yes
Strategy FE	No	Yes	No	Yes

- Above P90, floor trades pay 4.35¢ less and move futures less
- Below P25, neither venue moves futures, yet the floor costs 3.49¢ more

By Trade Size: Price Impact and Realized Spread

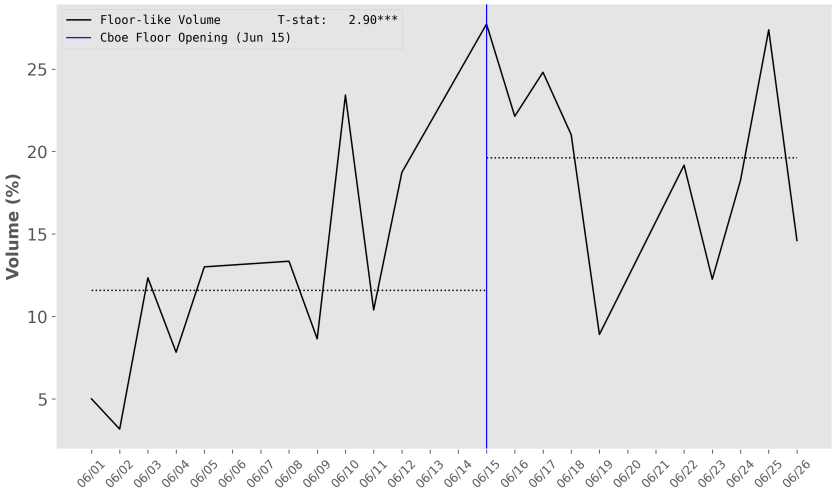
Return Futures Price Impact



- Up to P25: floor 7.77¢, almost all realized spread; electronic 1.90¢, AIM 1.72¢
- Above P90: price impact is 3.66¢ of 5.42¢ electronic (AIM 5.02¢ of 6.08¢) and −1.06¢ of 1.31¢ on the floor

Price impact and realized spread over one minute, in cents per unit of underlying

The Largest, Most Complex Trades Return With the Floor



- Their share is 11.6% in the two weeks before the opening and 19.6% after

Return

Replication Using AIM Trades Only

Return

Target: 90th-percentile floor trade = 1,950 contracts					
Trades Required	(mean / median)		36.2 / 30.0		
Elapsed Minutes	(mean / median)		4.1 / 3.3		
AIM Spread Paid	($)		6,166		
AIM Total Cost	($)		6,251–6,253		
		±1%	±2.5%	±5%	±10%
Floor Total Cost	($)	2,439	3,357	4,246	3,745
Floor Advantage	(%)	**61.0**	**46.3**	**32.1**	**40.1**

- AIM trades are larger, so fewer are needed, but they arrive less often
- The floor is 32–61% cheaper, close to the all-electronic replication

Priority on the Cboe Floor

Return

An order represented to the crowd executes in this sequence

1. The highest bid and the lowest offer trade first
2. At the same price, Priority Customer orders in the Book come next, in time priority
3. The DPM or LMM then receives 50%, 40%, or 30% of the remaining contracts
   With one, two, or three or more other quotes, if Cboe applies it to the class
4. In-crowd bids and offers follow in the order they were made
   Split equally when the sequence cannot be determined
5. Broker-dealer orders and quotes in the Book come last

Source: Cboe Rule Book effective October 7, 2019, Rule 5.85(a), pp. 340–341. Priority Customers are public customers who are not professionals. The Book is the electronic book of simple orders.

Small Floor Trades More Often Go to a Single Counterparty

Return

Fills Covering the Whole Leg (%)	AIM	Floor	Floor − AIM
All Trades	26.8	14.3	−12.5
≤ P25 (≤ 166)	29.9	**46.8**	**16.9**
P25–P50 (167–380)	16.3	27.2	10.9
P50–P75 (381–839)	12.4	13.6	1.2
P75–P90 (840–1,948)	7.0	3.4	−3.6
> P90 (≥ 1,949)	3.9	**1.3**	**−2.7**

175,051 AIM fills (Jun 1–12) and 28,083 floor fills (Jun 15–26), conditional on execution; electronic executions grouped into orders

- Up to P50, a floor fill covers the whole leg more often than an AIM fill
- Above P75 it does so less often, consistent with brokers sharing large orders with the crowd